

P2 Report

Reasoning:

The task requires choosing the most cost-efficient sorting algorithm based on different costs. Selection sort, Insertion sort and Bubble sort have different characteristics in terms of comparisons and swaps, so we need to estimate the cost of them before actual sorting. Accord to the material, we need only focus on the reads/writes performed on the input array.

The estimation uses a simplified cost model. It assumes worst-case behaviour.

For Selection sort,

We need to compare $[(n-1) + (n-2) + (n-3) + \dots + 1] = (n-1) * n/2$ times and we need to swap at most $(n-1)$ times.

Therefore,

```
int read_count = 2*(n-1) + n * (n - 1);  
int write_count = 2*(n-1);
```

For Insertion sort,

We use a temporary variable called temp_post to hold the first unsorted elements in the array. Therefore, we need to read $(n-1)$ times. Then, we need to compare and shift $[(n-1) + (n-2) + \dots + 1] = (n-1) * n/2$ times.

Therefore,

```
int read_count = (n-1) + n*(n-1);  
int write_count = (n-1) + n * (n-1)/2;
```

For Bubble sort,

We need to compare $(n-1)*n$ times and We need to swap $[(n-1) + (n-2) + \dots + 1] = (n-1)*n/2$ times.

Therefore,

```
int read_count = 2*n*(n-1) + n * (n - 1);  
int write_count = n*(n-1);
```

After estimation of each sorting method, we could know which one is the most efficient way. And then choose the most efficient way to deal with this task.

Evidence of Testing:

```
/Users/yihaoyu/Desktop/FIT2085/Ontrack_Files/P2/P2task
The cost for Selection sort is: 126
The cost for Insertion sort is: 270
The cost for Bubble sort is: 360
So I would choose: Selection sort
Cost of sorting was 126
# likes are:
100 96 92 88 84 80 76 72 68 64
```

```
/Users/yihaoyu/Desktop/FIT2085/Ontrack_Files/P2/P2task
The cost for Selection sort is: 396
The cost for Insertion sort is: 738
The cost for Bubble sort is: 1440
So I would choose: Selection sort
Cost of sorting was 396
# likes are:
9 8 7 6 5 4 3 2 1 0 |
```

```
/Users/yihaoyu/Desktop/FIT2085/Ontrack_Files/P2/P2task
The cost for Selection sort is: 2178
The cost for Insertion sort is: 2034
The cost for Bubble sort is: 5490
So I would choose: Insertion sort
Cost of sorting was 2034
# likes are:
9 8 7 6 5 4 3 2 1 0
```

Word Count: 203.